

1)

Relation: STUDENT_COURSE (SID, SName, CID, CName, Instructor, Grade)

Functional Dependencies (FDs):

1. $SID \rightarrow SName$
2. $CID \rightarrow CName$
3. $CID \rightarrow Instructor$
4. $(SID, CID) \rightarrow Grade$

Candidate Key Determination:

- $SID^+ = \{SID, SName\} \rightarrow$ Not a candidate key.
- $CID^+ = \{CID, CName, Instructor\} \rightarrow$ Not a candidate key.
- $(SID, CID)^+ = \{SID, SName, CID, CName, Instructor, Grade\} \rightarrow$ Determines all attributes.

Candidate Key:

(SID, CID)

- **SID** uniquely identifies the student name.
- **CID** uniquely identifies the course name and instructor.
- A student's **Grade** depends on both the student and the course.
- Therefore, only the combination **(SID, CID)** can uniquely identify every tuple in the relation, making it the **candidate key**.

Final Result:

- **Functional Dependencies:**
 - $SID \rightarrow SName$
 - $CID \rightarrow CName$
 - $CID \rightarrow Instructor$
 - $(SID, CID) \rightarrow Grade$
- **Candidate Key: (SID, CID)**

2)

Unnormalized Relation (UNF):

StudentID	StudentName	Course
S101	Rahul	DBMS, Java, Python
S102	Priya	C, C++, DBMS

Here, the **Course** attribute contains multiple values, violating 1NF.

Conversion to First Normal Form (1NF):

Step 1: Identify the repeating group (Course).

Step 2: Replace the multi-valued attribute with single (atomic) values.

StudentID	StudentName	Course
S101	Rahul	DBMS
S101	Rahul	Java
S101	Rahul	Python
S102	Priya	C
S102	Priya	C++
S102	Priya	DBMS

Justification:

- Each attribute contains only **one value** (atomic value).
- Repeating groups are eliminated.
- Every row is uniquely identified by the combination (**StudentID, Course**).
- Hence, the relation satisfies the **First Normal Form (1NF)**.

3)

Given:

- Relation: **R(A, B, C, D, E)**
- Functional Dependencies:
 - $A \rightarrow BC$
 - $BC \rightarrow D$
 - $D \rightarrow E$

Step 1: Identify the Candidate Key

- $A^+ = \{A, B, C, D, E\}$
- Therefore, **A** is the **candidate key**.

Step 2: Check for Partial Dependencies

- The candidate key consists of only one attribute (**A**).
- Since there is **no composite candidate key**, **partial dependency does not exist**.

Step 3: 2NF Analysis

- All non-prime attributes (**B, C, D, E**) are fully functionally dependent on the candidate key **A**.

- Hence, the relation already satisfies **Second Normal Form (2NF)**.

Result

No decomposition is required because the relation is already in **2NF**.

Final Relation:

R(A, B, C, D, E)

Conclusion:

Since the candidate key is a single attribute (**A**), there are no partial dependencies. Therefore, the given relation is already in **2NF**, and no decomposition is necessary.

4)

Given Relation:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

Functional Dependencies

- $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$
- $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

Transitive Dependency

- $\text{EmpID} \rightarrow \text{DeptID}$
- $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$
- Therefore, **$\text{EmpID} \rightarrow \text{DeptName}, \text{ManagerID}$** (Transitive Dependency)

3NF Decomposition

Relation 1: EMPLOYEE

- **EMPLOYEE(EmpID, EmpName, DeptID)**

Relation 2: DEPARTMENT

- **DEPARTMENT(DeptID, DeptName, ManagerID)**

Explanation

- The transitive dependency is removed by separating department details into a different relation.
- **EMPLOYEE** stores employee information and the department ID.
- **DEPARTMENT** stores department details and manager information.
- Both relations satisfy **Third Normal Form (3NF)** because all non-key attributes depend only on the primary key.

Final Result

- **EMPLOYEE(EmpID, EmpName, DeptID)**
- **DEPARTMENT(DeptID, DeptName, ManagerID)**

5)

Given:

- Relation: **R(A, B, C, D)**
- Functional Dependencies:
 - $AB \rightarrow C$
 - $C \rightarrow D$
 - $D \rightarrow A$

Step 1: Find the Candidate Key

- $(AB)^+ = \{A, B, C, D\}$
- Therefore, **AB** is the **candidate key**.

Step 2: Check BCNF

A relation is in **BCNF** if, for every functional dependency $X \rightarrow Y$, **X** is a super key.

Functional Dependency	Super Key?	BCNF Status
$AB \rightarrow C$	Yes	Satisfies BCNF
$C \rightarrow D$	No	Violates BCNF
$D \rightarrow A$	No	Violates BCNF

Hence, the relation is **not in BCNF**.

Step 3: BCNF Decomposition

Using $C \rightarrow D$:

- **$R_1(C, D)$**
- **$R_2(A, B, C)$**

The relation **$R_2(A, B, C)$** satisfies BCNF because **AB** is the candidate key and $AB \rightarrow C$.

Final Decomposed Relations

- **$R_1(C, D)$**
- **$R_2(A, B, C)$**

Conclusion

Since $C \rightarrow D$ and $D \rightarrow A$ violate the BCNF condition (their determinants are not super keys), the relation is decomposed into **$R_1(C, D)$** and **$R_2(A, B, C)$** , which satisfy **BCNF**.

6)

Given Relation:

Functional Dependency:

- $A \rightarrow B$

Decomposition:

- $R_1(A, B)$
- $R_2(A, C)$

Chase Table

Relation	A	B	C
$R_1(A, B)$	a	b	c_1
$R_2(A, C)$	a	b_1	c

Applying the FD $A \rightarrow B$:

- Since **A** has the same value (**a**) in both rows, **B** must also be the same.
- Replace **b₁** with **b**.

Updated Table

Relation	A	B	C
$R_1(A, B)$	a	b	c_1
$R_2(A, C)$	a	b	c

Verification

- One row contains only the original symbols (**a, b, c**).
- Therefore, the decomposition is **Lossless-Join**.

Conclusion

The decomposition $R(A, B, C) \rightarrow R_1(A, B)$ and $R_2(A, C)$ is **lossless** because applying the Chase Algorithm reconstructs the original relation without any loss of information.

7)

Given Relation:

TEACHER(TID, Subject, Hobby)

Multi-Valued Dependencies (MVDs)

- $TID \twoheadrightarrow Subject$
- $TID \twoheadrightarrow Hobby$

A teacher can teach multiple subjects and also have multiple hobbies. These attributes are independent of each other, causing redundancy.

4NF Decomposition

Relation 1:**TEACHER_SUBJECT(TID, Subject)**

TID	Subject
T1	DBMS
T1	Java
T2	Python

Relation 2:**TEACHER_HOBBY(TID, Hobby)**

TID	Hobby
T1	Reading
T1	Music
T2	Painting

Explanation

- The independent multi-valued attributes **Subject** and **Hobby** are separated into two relations.
- This removes redundancy and prevents unnecessary repetition of data.
- Each relation contains only one multi-valued dependency and satisfies **Fourth Normal Form (4NF)**.

Final Result

- **TEACHER_SUBJECT(TID, Subject)**
- **TEACHER_HOBBY(TID, Hobby)**

8)

Consider the relation:

SUPPLY(SupplierID, PartID, ProjectID)

A supplier can supply multiple parts to multiple projects. Storing all three attributes in a single relation creates redundancy due to the join dependency.

Join Dependency

The relation satisfies the join dependency:

SUPPLY = SUPPLIER_PART ⋈ PART_PROJECT ⋈ SUPPLIER_PROJECT**5NF Decomposition****Relation 1:****SUPPLIER_PART(SupplierID, PartID)**

SupplierID	PartID
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S1	P1
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S1	P2
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S2	P3
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Relation 2:

PART_PROJECT(PartID, ProjectID)

PartID	ProjectID
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P1	PR1
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P2	PR2
----	-----

P3	PR3
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Relation 3:

SUPPLIER_PROJECT(SupplierID, ProjectID)

SupplierID	ProjectID
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S1	PR1
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S1	PR2
----	-----

S2	PR3
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Explanation

- The original relation contains redundant tuples because of join dependencies.
- Decomposing it into **SUPPLIER_PART**, **PART_PROJECT**, and **SUPPLIER_PROJECT** eliminates redundancy.
- The original relation can be reconstructed using a **natural join** of the three relations.
- Hence, the decomposed relations satisfy **Fifth Normal Form (5NF)**.

Final Result

- **SUPPLIER_PART(SupplierID, PartID)**
- **PART_PROJECT(PartID, ProjectID)**
- **SUPPLIER_PROJECT(SupplierID, ProjectID)**

9)

Poorly Designed Relation:

COLLEGE(StudentID, StudentName, DeptID, DeptName, HOD)

Anomalies

1. Insertion Anomaly:

- A new department cannot be added unless a student is enrolled in it.

2. Update Anomaly:

- If the department name or HOD changes, it must be updated in multiple records, leading to inconsistency.

3. Deletion Anomaly:

- Deleting the last student of a department also deletes the department information.

Functional Dependencies

- $\text{StudentID} \rightarrow \text{StudentName}, \text{DeptID}$
- $\text{DeptID} \rightarrow \text{DeptName}, \text{HOD}$

3NF Decomposition

STUDENT(StudentID, StudentName, DeptID)

StudentID	StudentName	DeptID
S101	Rahul	D01
S102	Priya	D02

DEPARTMENT(DeptID, DeptName, HOD)

DeptID	DeptName	HOD
D01	CSE	Dr. Kumar
D02	ECE	Dr. Meena

Conclusion

By decomposing the relation into **STUDENT** and **DEPARTMENT**, insertion, update, and deletion anomalies are removed. Both relations satisfy **Third Normal Form (3NF)**.

10)

Given Functional Dependencies:

- $A \rightarrow BC$
- $BC \rightarrow D$
- $D \rightarrow E$

Step 1: Compute Attribute Closure

Closure of A (A^+):

- $A \rightarrow BC \Rightarrow \{A, B, C\}$
- $BC \rightarrow D \Rightarrow \{A, B, C, D\}$

- $D \rightarrow E \Rightarrow \{A, B, C, D, E\}$

Therefore,

$$A^+ = \{A, B, C, D, E\}$$

Step 2: Determine the Candidate Key

Since A^+ contains all the attributes of the relation, **A** is a **candidate key**.

Verification

- $B^+ = \{B\}$
- $C^+ = \{C\}$
- $D^+ = \{D, E\}$
- $E^+ = \{E\}$

None of these determine all the attributes.

Final Answer

- **Attribute Closure:**
 $A^+ = \{A, B, C, D, E\}$
- **Candidate Key:**
A

Conclusion:

The closure of **A** includes all attributes in the relation. Hence, **A** is the **only candidate key** for the given functional dependencies.